

Two-Point Correlator Analysis

Two_pt_ML_OTSP_D_gold_to_nongold_Experiment

Analysis Configuration

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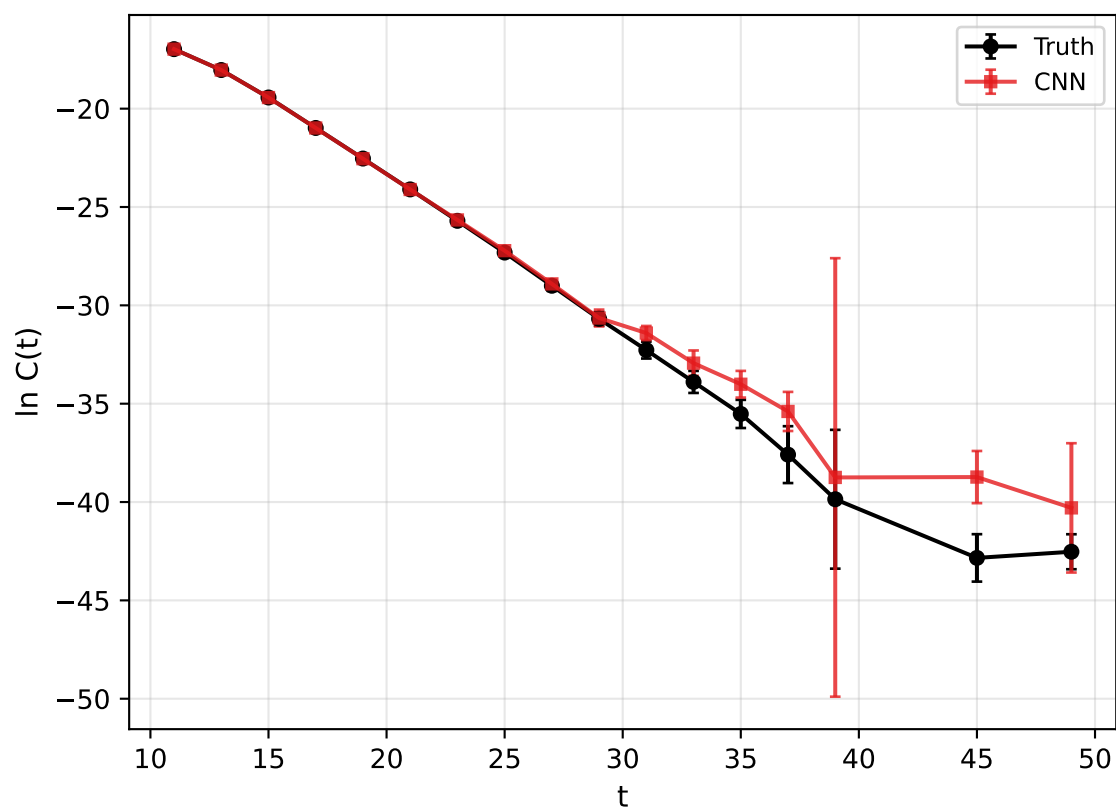
Binning: 4 samples/bin -> 307 bins
Fit range: tmin = 14 (truth), 22 (ML)
 tmax = 40
SVD cut: 0.01
Prior width: 0.4
States: 3 normal + 2 oscillating
Time extent: 95

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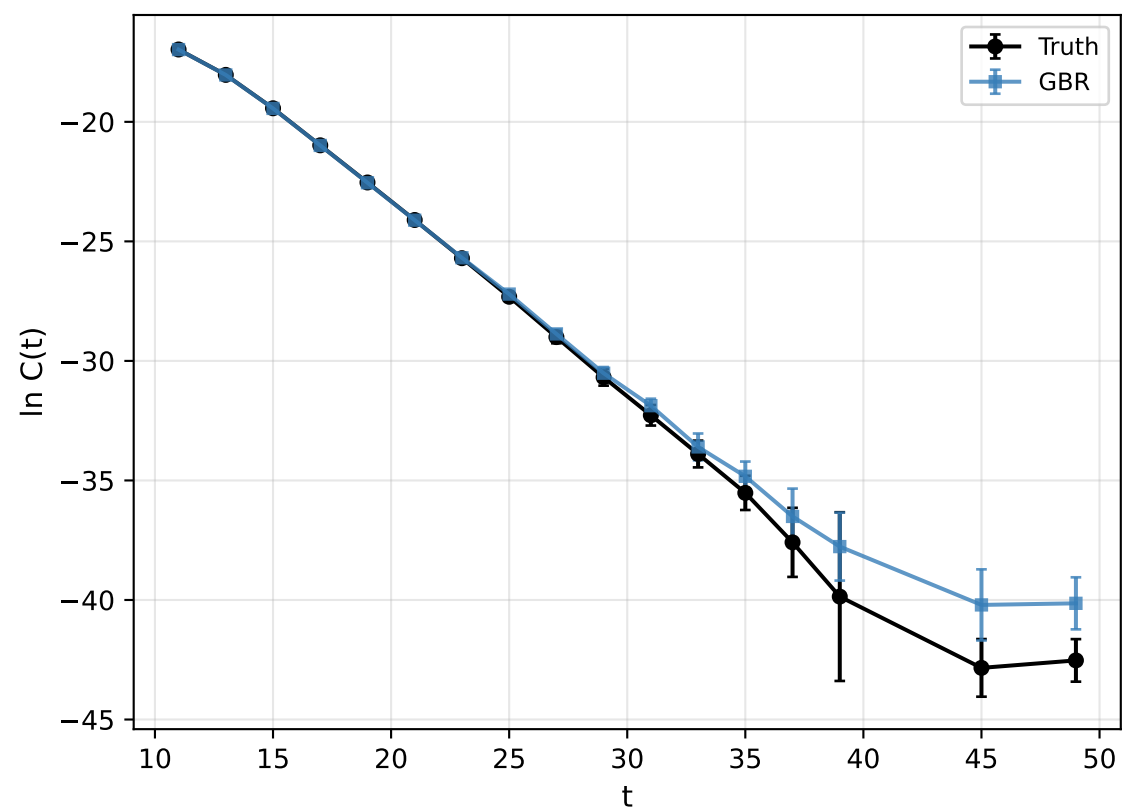
ML Models: CNN, GBR, MLP, Transformer
Methods: Ratio Method, Bias Corrected

Log Correlator vs Time: ML (Ratio Method) vs Truth

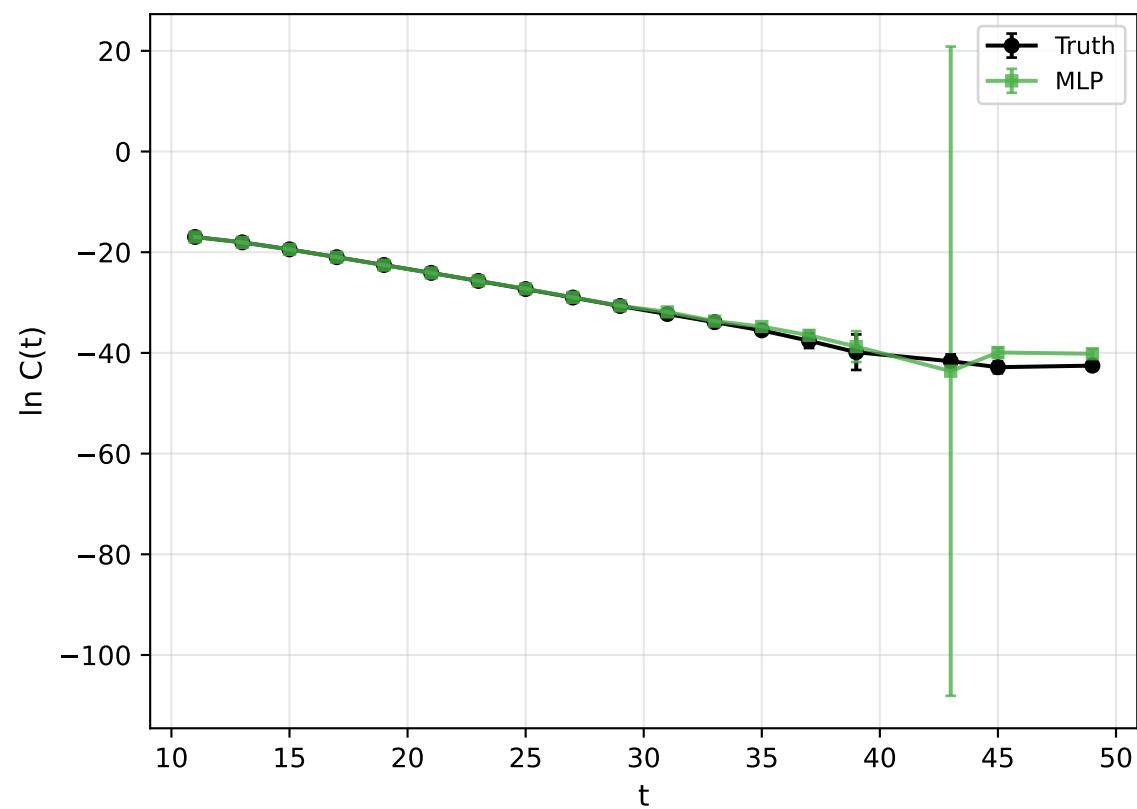
CNN vs Truth



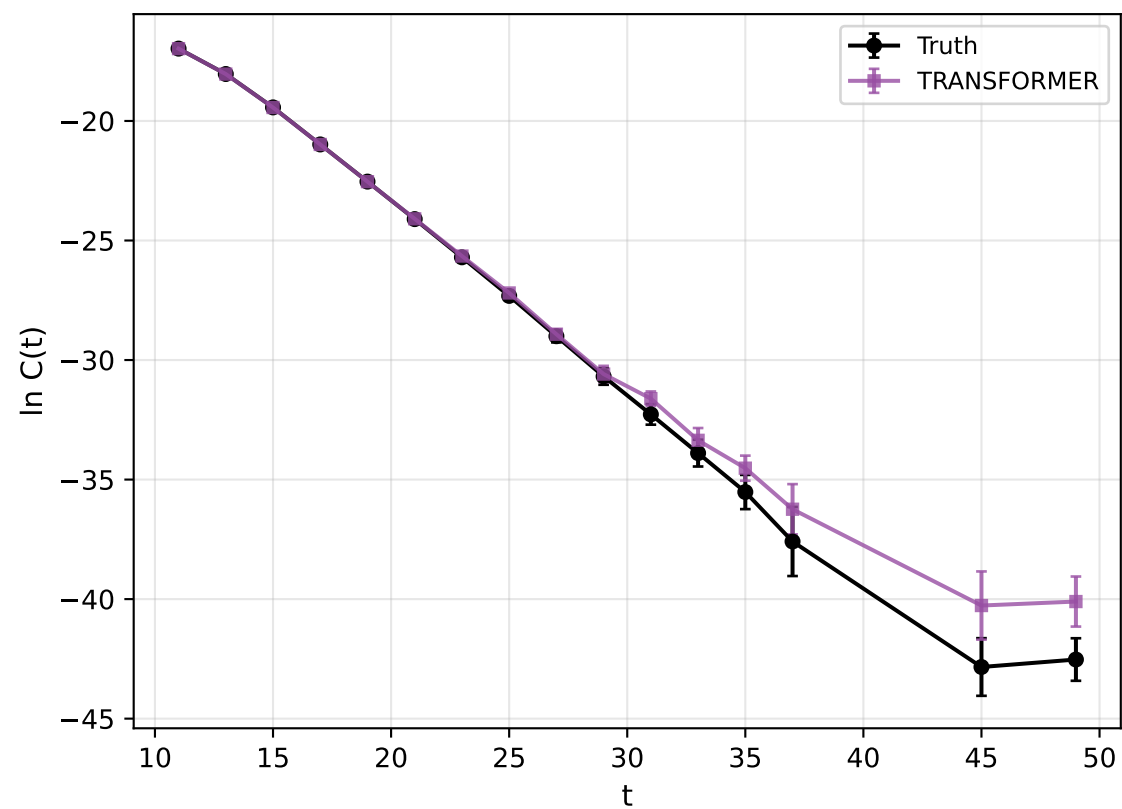
GBR vs Truth



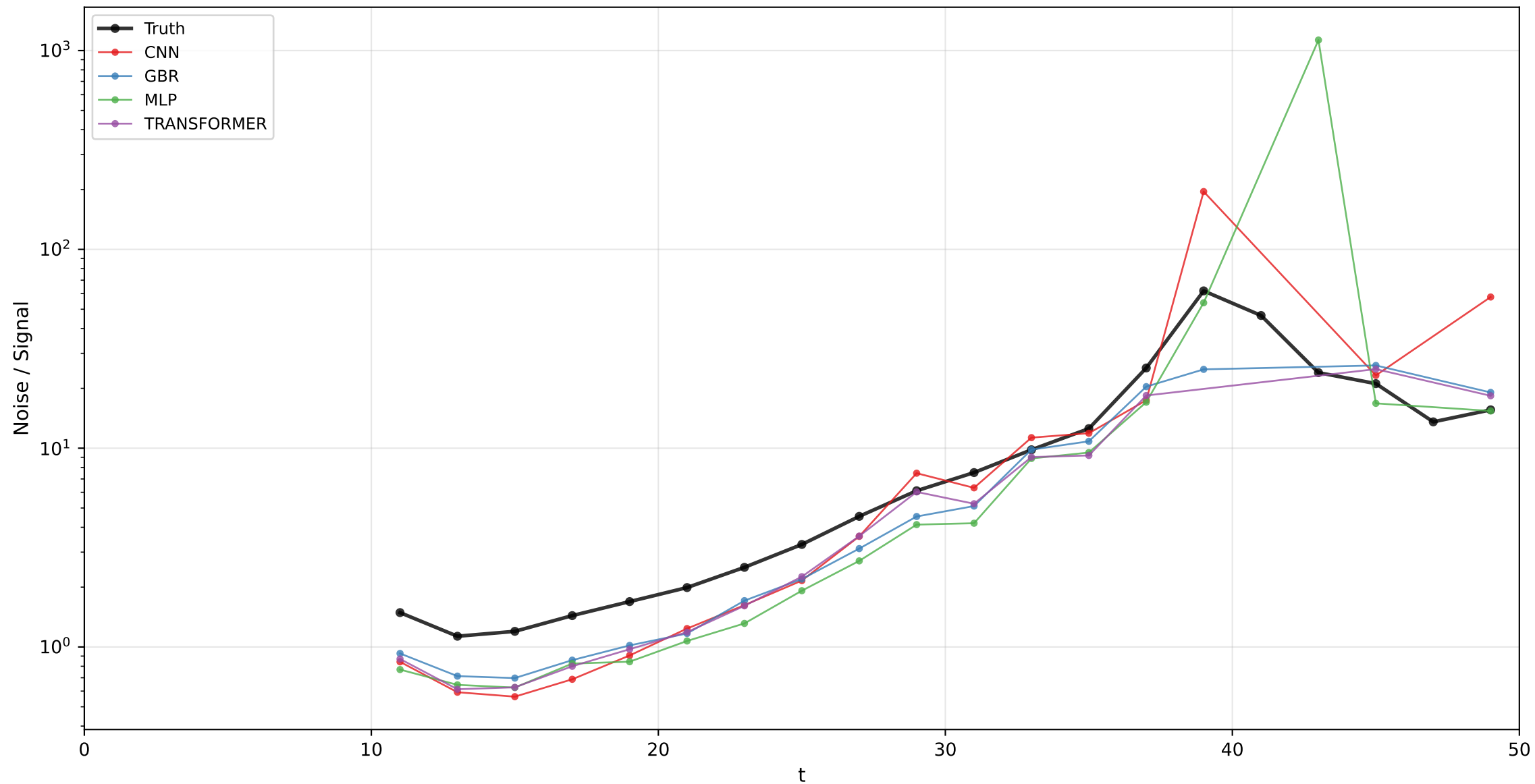
MLP vs Truth



TRANSFORMER vs Truth

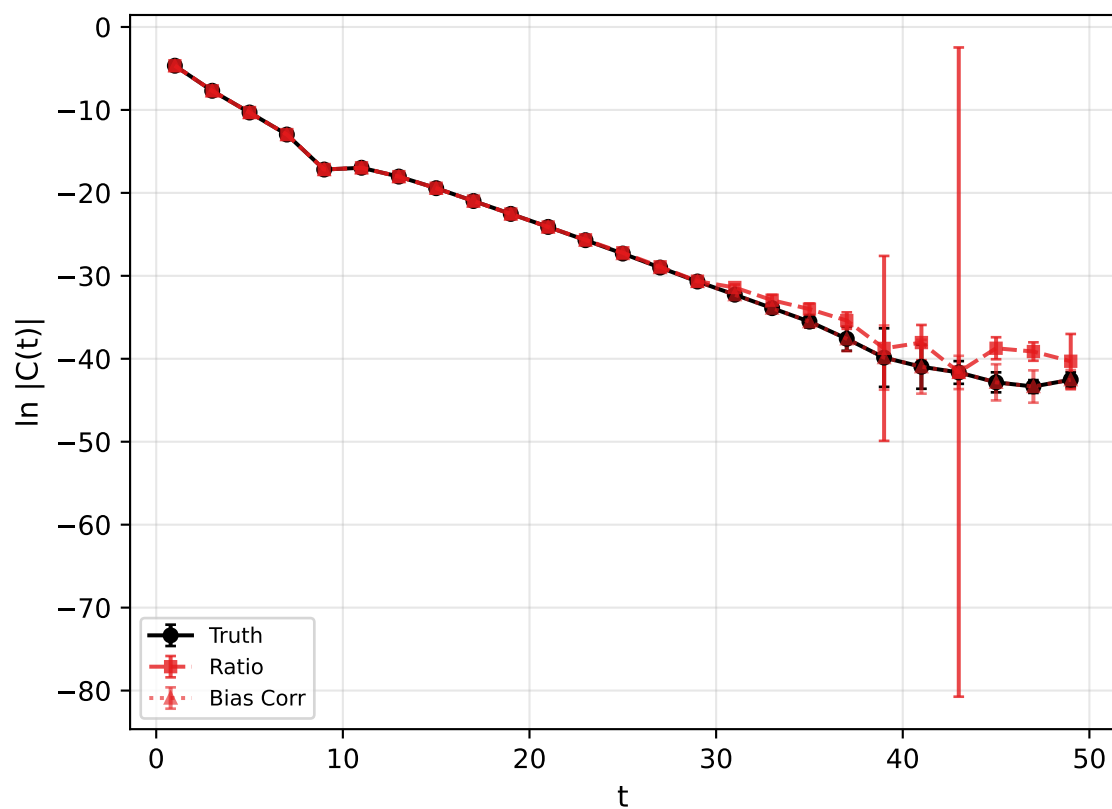


Noise-to-Signal Ratio vs Time (Ratio Method)

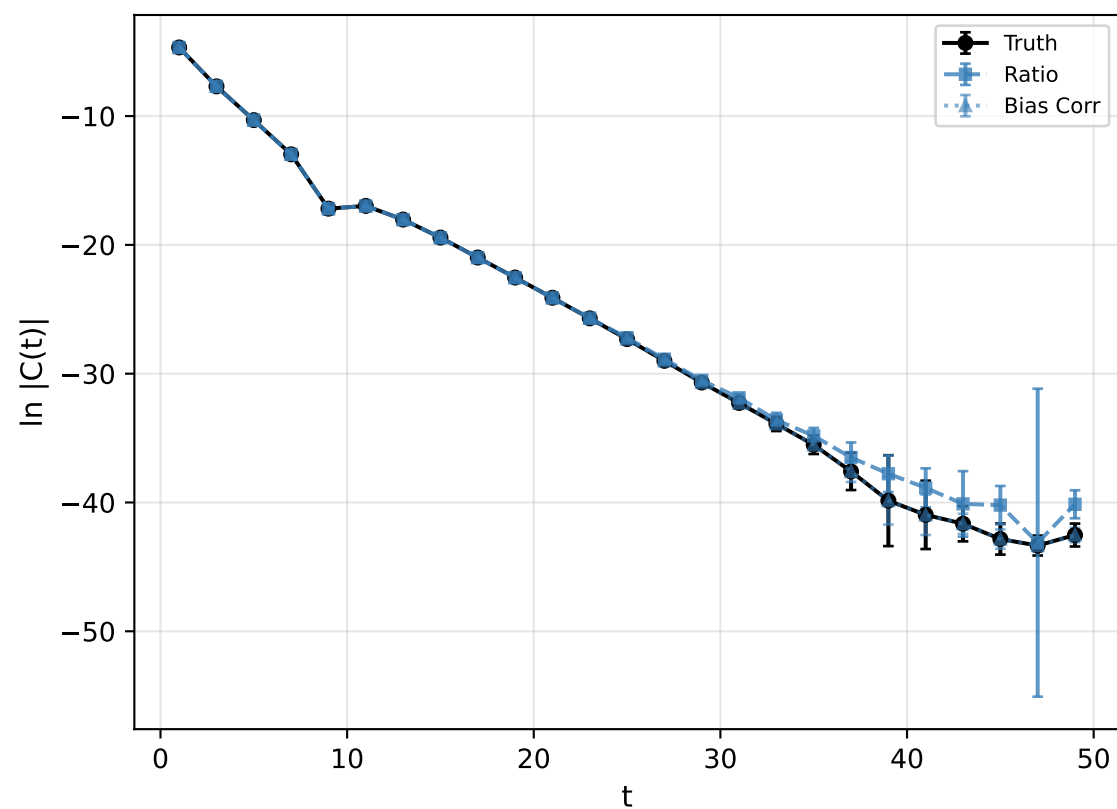


Log Correlator: Truth vs Ratio Method vs Bias Corrected

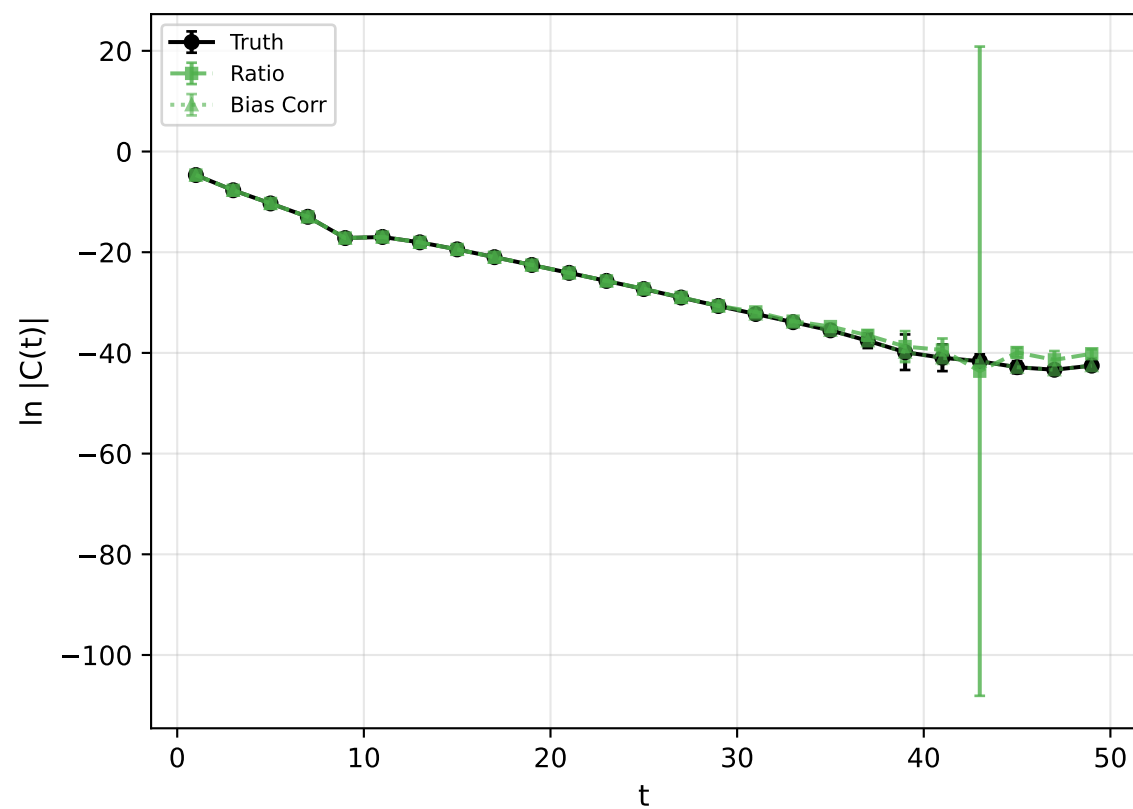
CNN



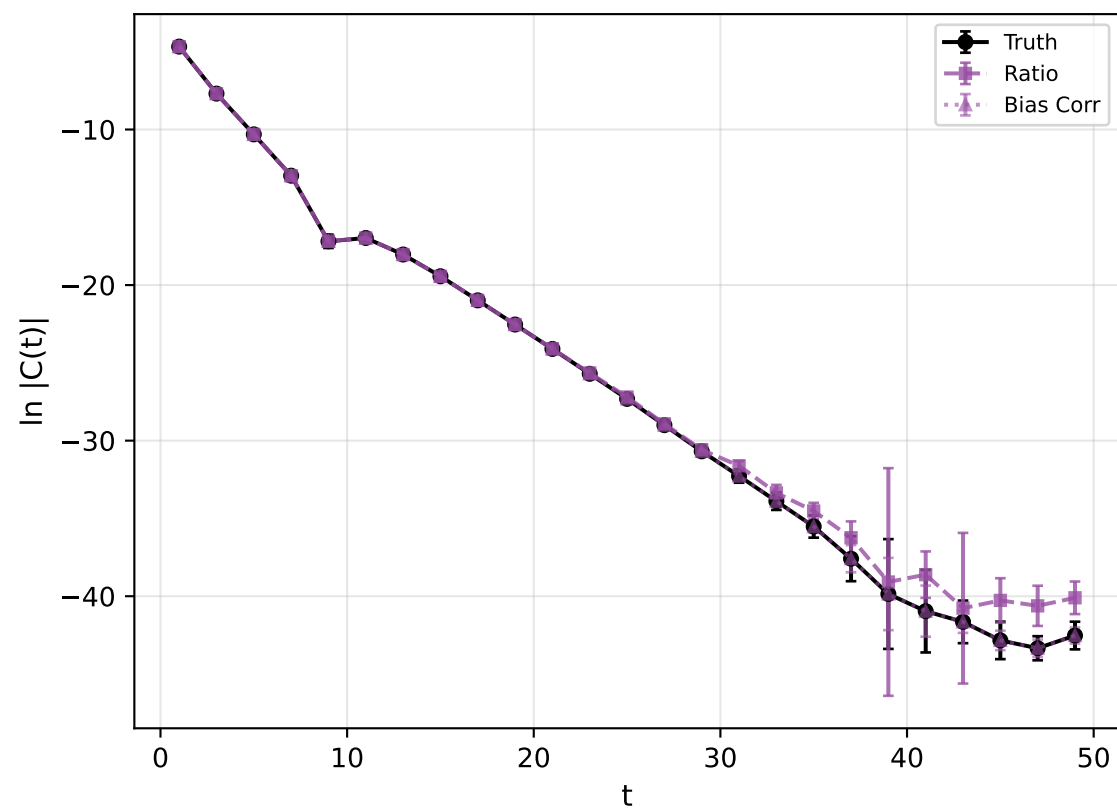
GBR



MLP



TRANSFORMER



Fit Results Summary

Model	Method	dE0	a0	chi2/dof	Q
Truth	-	0.8354(0.0067)	0.0455(0.0026)	0.851	0.6860
CNN	Ratio	0.8224(0.0108)	0.0402(0.0049)	1.163	0.2795
	Bias Corr	0.8312(0.0075)	0.0439(0.0038)	0.844	0.6547
GBR	Ratio	0.8133(0.0120)	0.0350(0.0049)	0.850	0.6473
	Bias Corr	0.8214(0.0090)	0.0382(0.0040)	0.742	0.7784
MLP	Ratio	0.8089(0.0099)	0.0331(0.0038)	1.209	0.2384
	Bias Corr	0.8213(0.0076)	0.0379(0.0033)	1.721	0.0260
TRANSFORMER	Ratio	0.8176(0.0131)	0.0376(0.0056)	0.743	0.7769
	Bias Corr	0.8268(0.0089)	0.0408(0.0041)	0.572	0.9283

Fit Parameters: Ratio Method vs Bias Corrected

